

First **Postulate** of QM

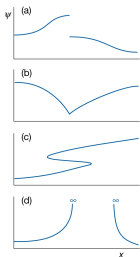
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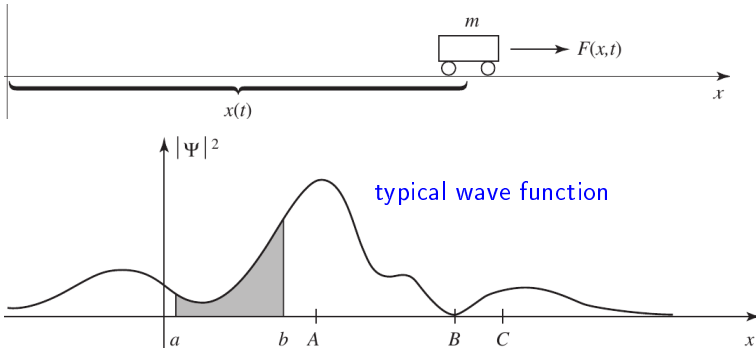
$$|\psi(\vec{r}, t)|^2 d\vec{r} = \text{Probability of locating system in vol. } \vec{r} \rightarrow \vec{r} + d\vec{r}$$

$$\text{in vol. } \Delta V, \text{ Probability of locating system} = \int_{\Delta V} |\psi(\vec{r}, t)|^2 d\vec{r}$$

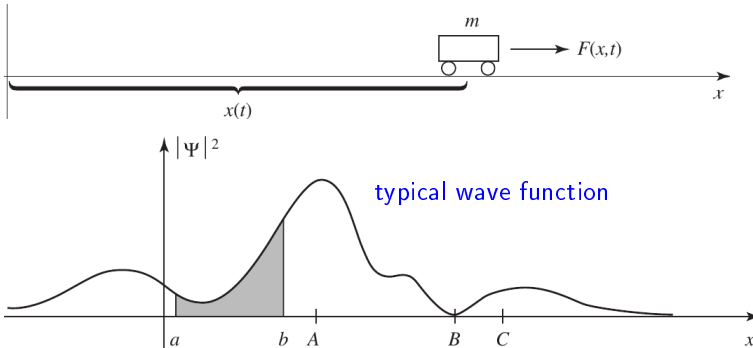
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Probability of locating the particle in the shaded region = $\int_a^b |\psi(x)|^2 dx$

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$$\therefore \text{probability} = 4 \times (19.85)^2 \times 0.1 \times e^{-2 \times 19.85} = 9.04 \times 10^{-16}$$

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$$\langle\psi|\psi\rangle = \int |\psi(\vec{r}, t)|^2 d\tau = \textbf{Total Probability}$$

- usually normalized to 1 for a single particle

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standard basis in $\mathbb{C}^2$: $\{|0\rangle; |1\rangle\}$

Ex. : Verify orthonormality of the above unit vectors

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write as $|\psi'\rangle = \hat{A}|\psi\rangle$

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Hermitian? - next class

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P3. If $\Psi(q, t)$ is normalized state function of a system at time t , then average value of a physical observable B at time t is given by the **expectation** value

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Separation of variables and time independent SE

Assume : $\Psi(x, t) = \psi(x)\phi(t)$ (1-D for simplicity)

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$$\Rightarrow i\hbar \dot{\phi}(t) = E\phi(t); \text{ and } \boxed{\hat{H}\psi(x) = E\psi(x)} : \text{time independent SE};$$

an **eigenvalue equation**, depends on specific $V(x)$

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: **stationary states** - **states of definite total energy**

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resolution of paradox : $\Psi(x, t) = \psi_E(x)e^{-\frac{i}{\hbar}Et}$ is a 'particular' soln. of TDSE

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third term $\implies$ [interference](#)

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nontrivial time dependence $\Leftrightarrow$

after solving TISE one must "reconstitute" the wavepacket
- superpose solns. of TISE, each with its own amplitude and phase factor

Some elementary problems

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Thus ϕ is also a solution

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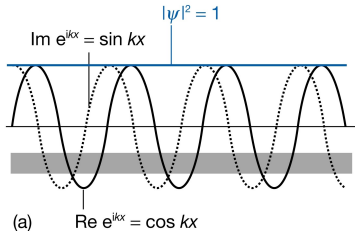
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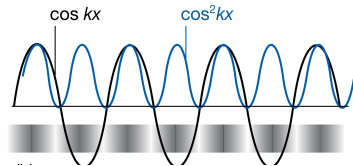
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Free particle with constant momentum



Probability distribution of state prepared by superposition of states of equal magnitude of momentum but opposite signs



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earlier expressions can be seen as superpositions of these new solutions.

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these **plane waves** solutions, are not useful for real systems
need to have particles confined in definite boundaries in space

More generally, allowing k to be +ve or -ve,

time dependent soln. : $\Psi_k(x, t) = Ae^{i(kx - \frac{\hbar k^2}{2m}t)}$; $k = \pm \frac{2mE}{\hbar}$, with

$$k > 0 \implies |\rightarrow\rangle$$

$$k < 0 \implies |\leftarrow\rangle$$

Compare with classical wave eqn. $\Psi_k(x, t) = Ae^{ik(x-vt)} \implies v = \frac{\hbar k}{2m} = \sqrt{\frac{E}{2m}}$:

phase velocity $\equiv \frac{\omega}{k}$

Classical speed of a free particle : $v = \sqrt{\frac{2E}{m}} \equiv \frac{\partial \omega}{\partial k}$: group velocity (of a wave packet)

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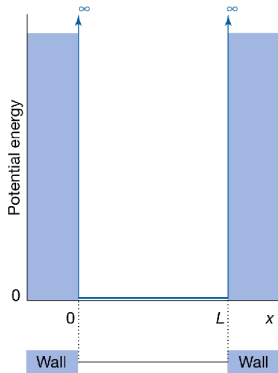
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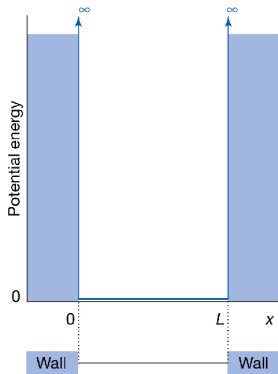
$$v_{\text{classical}} = v_{\text{group}} = 2v_{\text{phase}}$$

particle in 1-D box with

particle in 1-D box with $V(x) = 0$, $\forall 0 < x < L$ and ∞ elsewhere



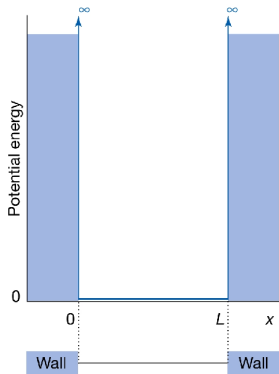
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infinite square well

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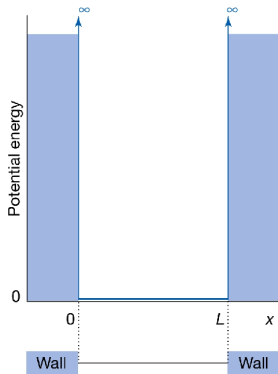


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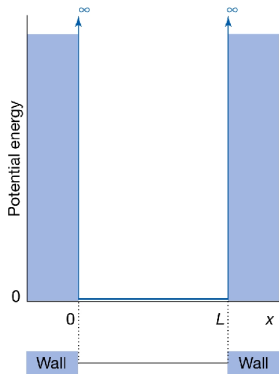
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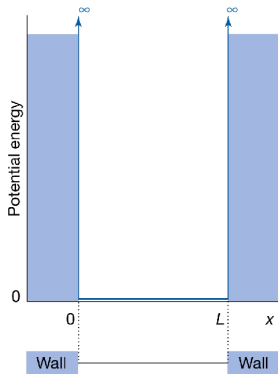
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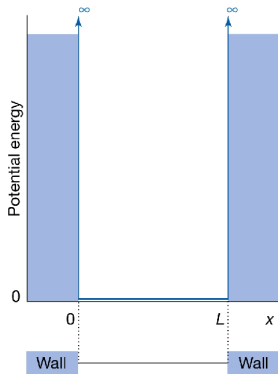
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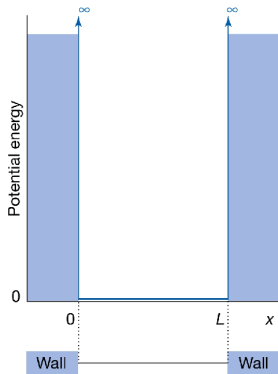
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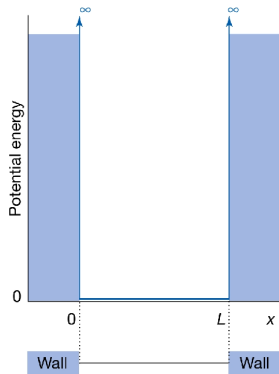
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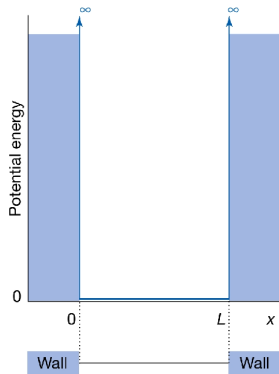
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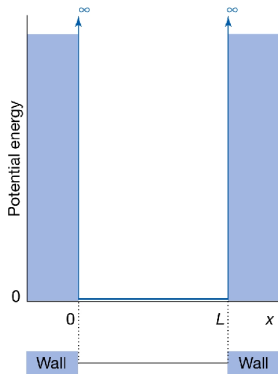
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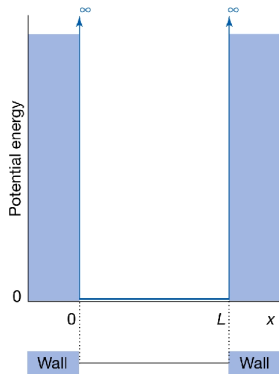
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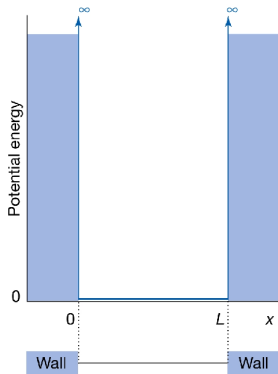
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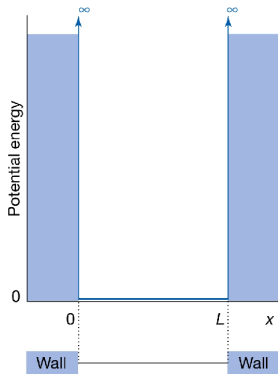
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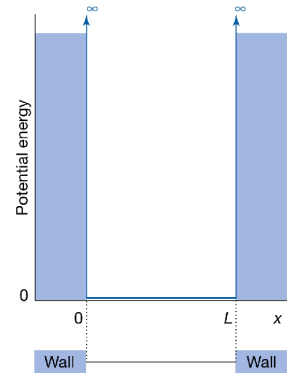
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Energy quantized,



also called

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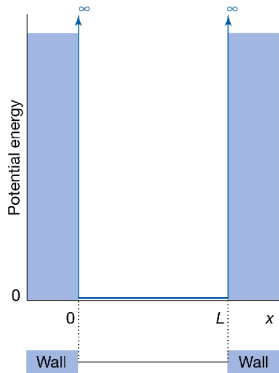
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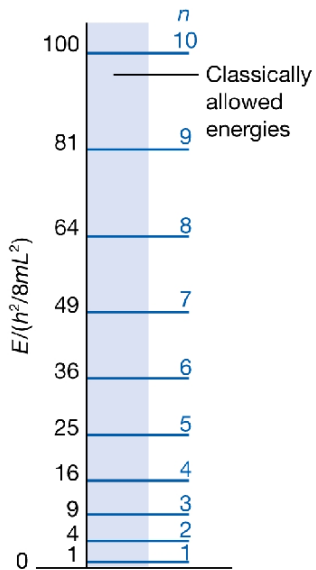
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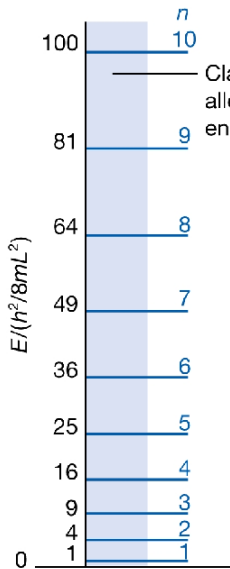
Energy quantized, $E_n = \frac{\hbar^2 k^2}{2m} = \frac{n^2 h^2}{8mL^2}$



also called

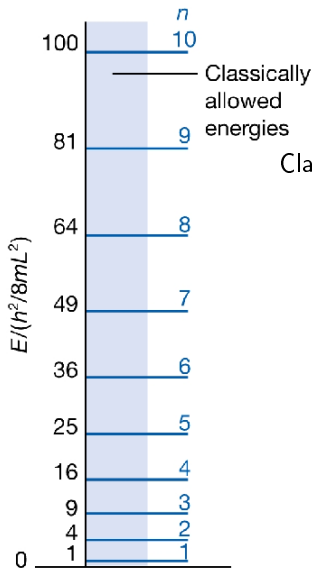
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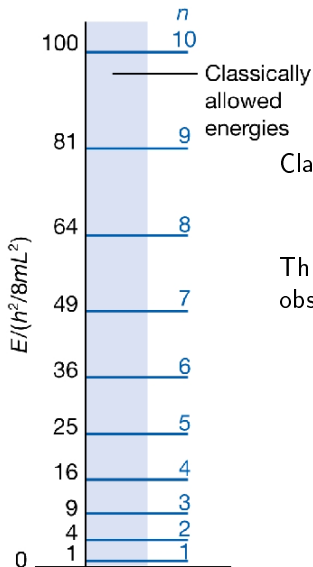
Classically
allowed
energies

Classically all energies are allowed continually



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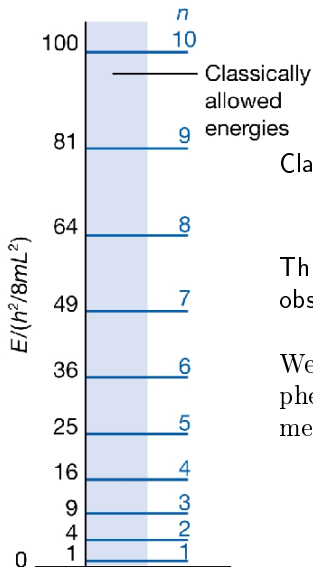
$$\text{QM} : E_n = \frac{n^2 h^2}{8mL^2}$$



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This expression is helpful in explaining several observations

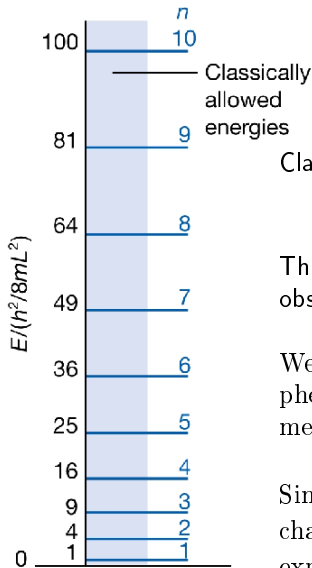


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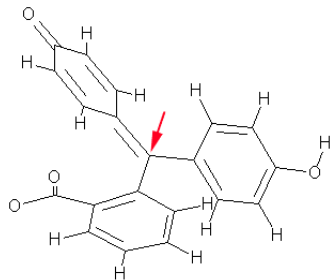
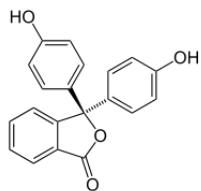
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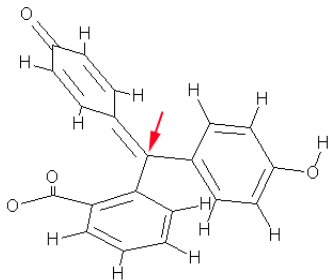
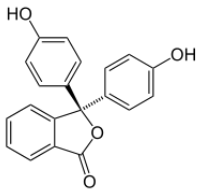
Similarly, change in color of a nano particle on changing its shape and size can be simply explained using this expression

Ex.1 :

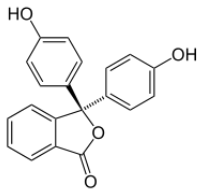


Ex.1 :

acidic medium : π electrons localised in three separate phenyl rings

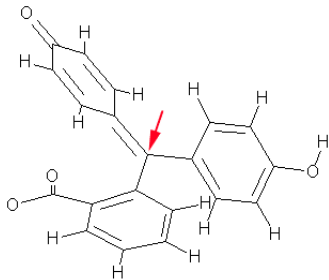


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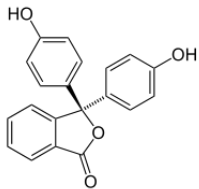


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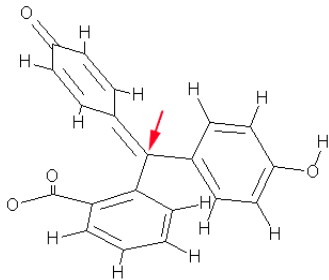
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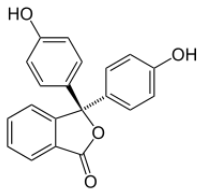
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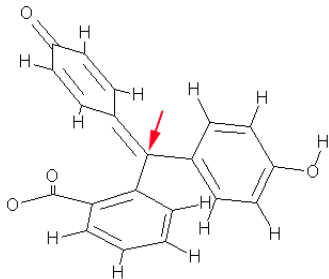
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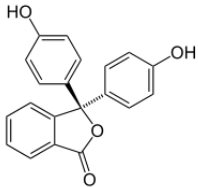
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$\therefore$ HOMO - LUMO energy gap larger in basic medium as compared to that in acidic medium

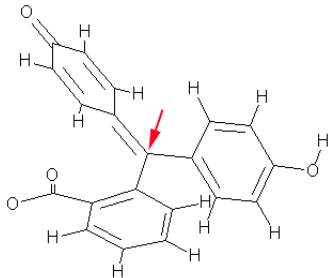
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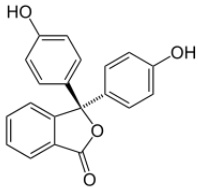
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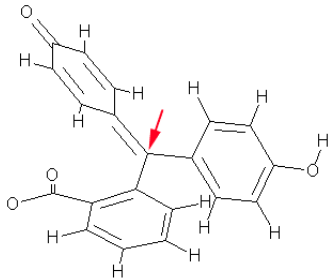
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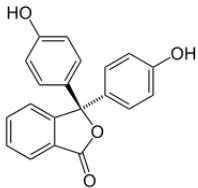
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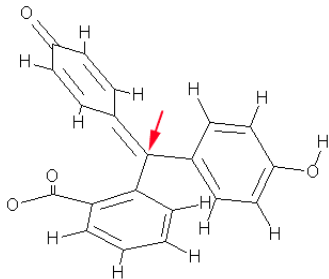
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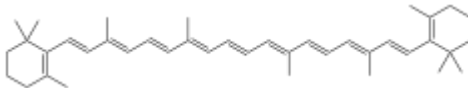
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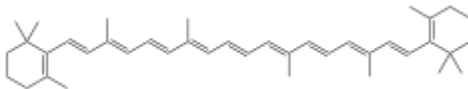
acidic medium : uv

Ex. 2 :

Ex. 2 : estimation of of C-C bondlength in conjugated part of β -carotene

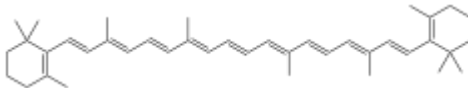


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absorption spectrum has a peak around 470 nm

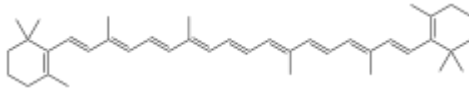
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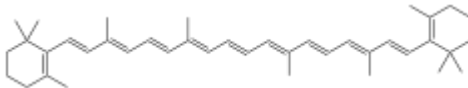
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calculate L from here and estimate average C-C bond length.

This comes very close to experimental value

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but we know that $\psi(x) \neq 0$ only in subdomain $0 < x < L$

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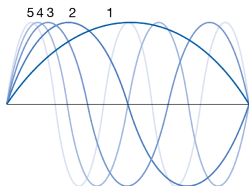
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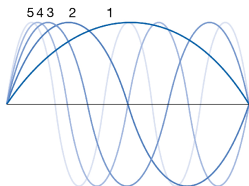
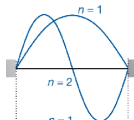
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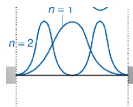
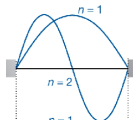
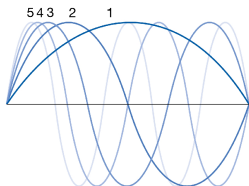
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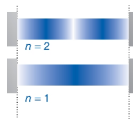
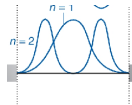
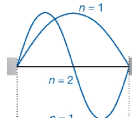
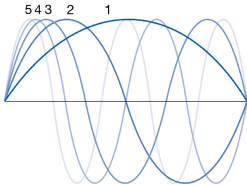
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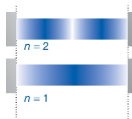
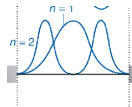
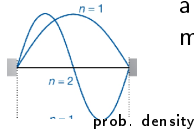
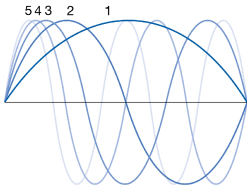
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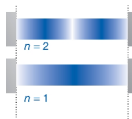
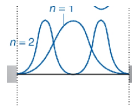
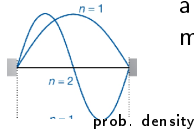
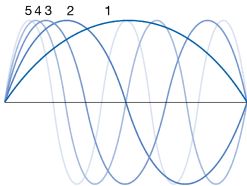
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$n - 1$ nodes : function change
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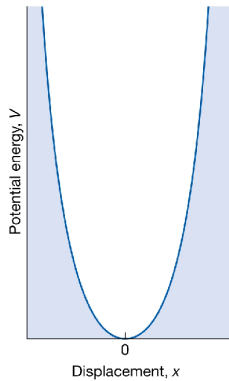
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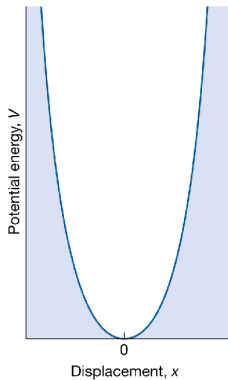
without this crucial feature SE would be incompatible with the statistical interpretation, and the whole theory would crumble

1-D HO : $V(x) = \frac{1}{2}kx^2$; $k = \mu\omega^2$

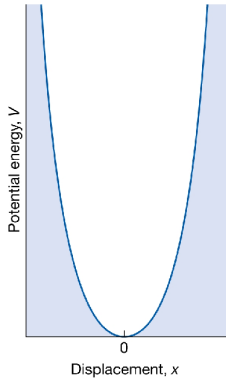


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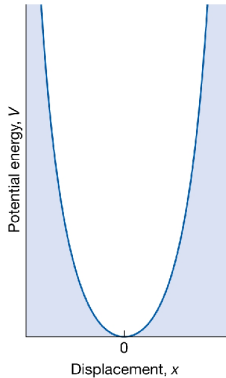
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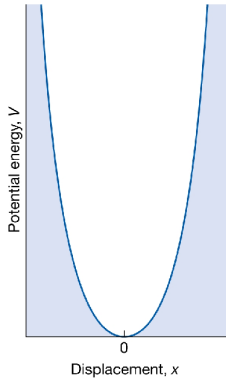


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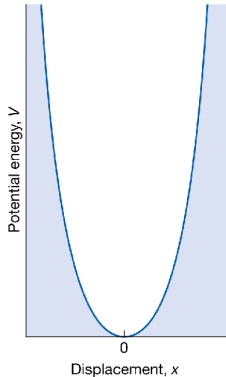
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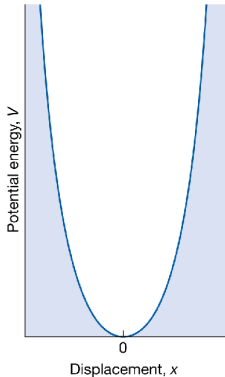
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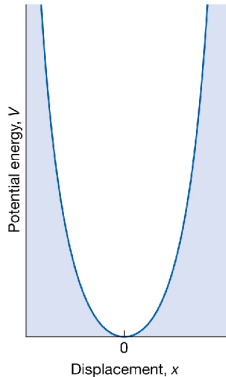
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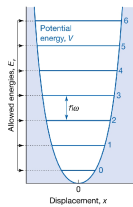
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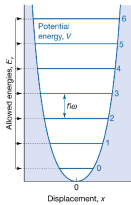
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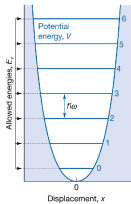
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energy states :

$$E_v = (v + \frac{1}{2})h\nu, \text{ where } \nu = \omega/2\pi = \frac{1}{2\pi}\sqrt{\frac{k}{\mu}}$$

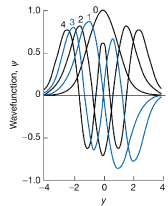


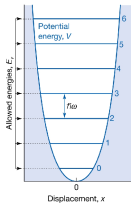


Notice : quantum number starts from 0

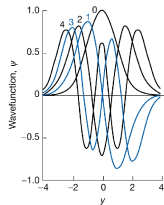


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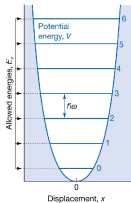




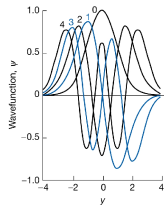
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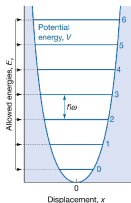
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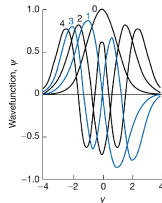
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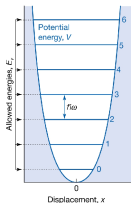
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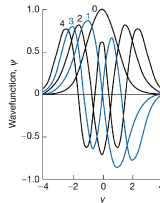
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Since there is no hard wall here, solution remains non-zero outside the energy barrier even if total energy is lower than PE



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It is called tunneling and has profound implication in proton transfer across barriers or electrons moving across junctions in semiconductor diodes

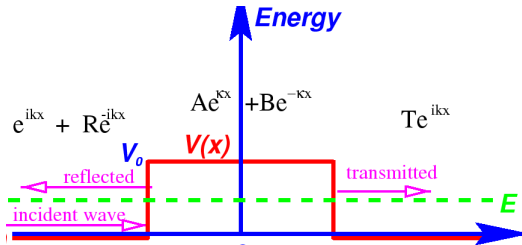
Quantum mechanical tunneling

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Problem: Find the probability that a particle of mass μ and energy E will penetrate a barrier of height $V(> E)$ and width L when it is incident from the left as shown :

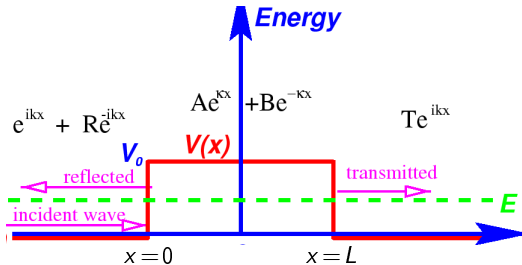
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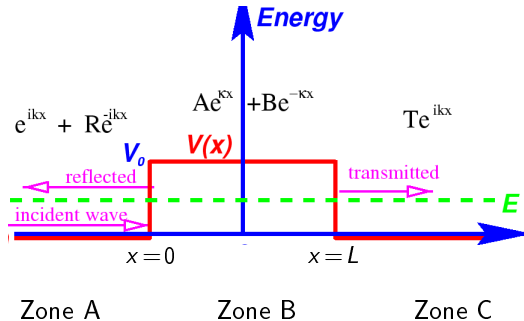
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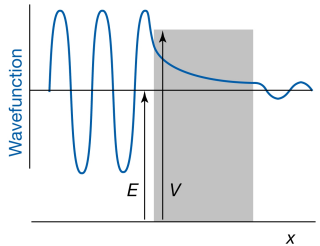
Zone A : $-\frac{\hbar^2}{2\mu}\psi_A''(x) \stackrel{V=0}{=} E\psi_A(x)$ or, $\psi_A''(x) + k^2\psi_A(x) = 0$; $\frac{\hbar^2 k^2}{2\mu} = E$

$$\psi_A(x) = \underbrace{Ae^{ikx}}_{\text{rightward moving}} + \underbrace{A'e^{-ikx}}_{\text{leftward moving (reflected)}}; \quad k = \sqrt{\frac{2\mu E}{\hbar^2}}$$

Zone B : $-\frac{\hbar^2}{2\mu}\psi_B''(x) + V\psi_B(x) = E\psi_B(x)$
 or, $\psi_B''(x) - \kappa^2\psi_B(x) = 0$; $\frac{\hbar^2 \kappa^2}{2\mu} = E - V$

$$\psi_B(x) = Be^{\kappa x} + B'e^{-\kappa x}; \quad \kappa = \sqrt{\frac{2\mu(V-E)}{\hbar^2}}$$

Zone C : $\psi_C(x) = \underbrace{Ce^{ikx}}_{\text{rightward moving (transmitted)}}$



$$\text{Zone A : } -\frac{\hbar^2}{2\mu}\psi_A''(x) \stackrel{V=0}{=} E\psi_A(x) \quad \text{or, } \psi_A''(x) + k^2\psi_A(x) = 0; \quad \frac{\hbar^2 k^2}{2\mu} = E$$

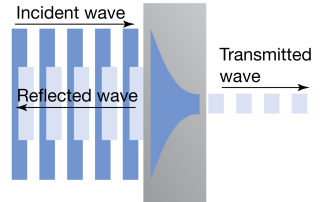
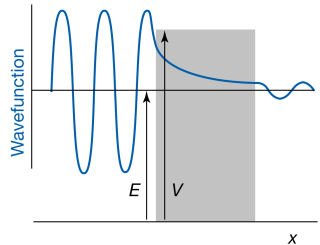
$$\psi_A(x) = \underbrace{Ae^{ikx}}_{\text{rightward moving}} + \underbrace{A'e^{-ikx}}_{\text{leftward moving (reflected)}}; \quad k = \sqrt{\frac{2\mu E}{\hbar^2}}$$

$$\text{Zone B : } -\frac{\hbar^2}{2\mu}\psi_B''(x) + V\psi_B(x) = E\psi_B(x)$$

$$\text{or, } \psi_B''(x) - \kappa^2\psi_B(x) = 0; \quad \frac{\hbar^2 \kappa^2}{2\mu} = E - V$$

$$\psi_B(x) = Be^{\kappa x} + B'e^{-\kappa x}; \quad \kappa = \sqrt{\frac{2\mu(V-E)}{\hbar^2}}$$

$$\text{Zone C : } \psi_C(x) = \underbrace{Ce^{ikx}}_{\text{rightward moving (transmitted)}}$$



$$\psi_A(x) = Ae^{ikx} + A'e^{-ikx}$$

$$\psi_B(x) = Be^{\kappa x} + B'e^{-\kappa x}$$

$$\psi_C(x) = Ce^{ikx}$$

$$\psi_A(x) = Ae^{ikx} + A'e^{-ikx}$$

$$\psi_B(x) = Be^{\kappa x} + B'e^{-\kappa x}$$

$$\psi_C(x) = Ce^{ikx}$$

using B.C. : $\psi_A(0) = \psi_B(0)$; $\psi'_A(0) = \psi'_B(0)$

$$\psi_A(x) = Ae^{ikx} + A'e^{-ikx}$$

$$\psi_B(x) = Be^{\kappa x} + B'e^{-\kappa x}$$

$$\psi_C(x) = Ce^{ikx}$$

using B.C. : $\psi_A(0) = \psi_B(0)$; $\psi'_A(0) = \psi'_B(0)$
we get, $A + A' = B + B'$

$$\psi_A(x) = Ae^{ikx} + A'e^{-ikx}$$

$$\psi_B(x) = Be^{\kappa x} + B'e^{-\kappa x}$$

$$\psi_C(x) = Ce^{ikx}$$

using B.C. : $\psi_A(0) = \psi_B(0)$; $\psi'_A(0) = \psi'_B(0)$

we get, $A + A' = B + B'$

and $(\alpha = ik)$, $\alpha A - \alpha A' = \kappa B - \kappa B'$

$$\psi_A(x) = Ae^{ikx} + A'e^{-ikx}$$

$$\psi_B(x) = Be^{\kappa x} + B'e^{-\kappa x}$$

$$\psi_C(x) = Ce^{ikx}$$

using B.C. : $\psi_A(0) = \psi_B(0)$; $\psi'_A(0) = \psi'_B(0)$

we get, $A + A' = B + B'$

and $(\alpha = ik)$, $\alpha A - \alpha A' = \kappa B - \kappa B'$

or, $2\alpha A = (\alpha + \kappa) B + (\alpha - \kappa) B'$

$$\psi_A(x) = Ae^{ikx} + A'e^{-ikx}$$

$$\psi_B(x) = Be^{\kappa x} + B'e^{-\kappa x}$$

$$\psi_C(x) = Ce^{ikx}$$

using B.C. : $\psi_A(0) = \psi_B(0)$; $\psi'_A(0) = \psi'_B(0)$

we get, $A + A' = B + B'$

and $(\alpha = ik)$, $\alpha A - \alpha A' = \kappa B - \kappa B'$

or, $2\alpha A = (\alpha + \kappa) B + (\alpha - \kappa) B'$

also, $\psi_C(L) = \psi_B(L)$; $\psi'_C(L) = \psi'_B(L)$

$$\psi_A(x) = Ae^{ikx} + A'e^{-ikx}$$

$$\psi_B(x) = Be^{\kappa x} + B'e^{-\kappa x}$$

$$\psi_C(x) = Ce^{ikx}$$

using B.C. : $\psi_A(0) = \psi_B(0)$; $\psi'_A(0) = \psi'_B(0)$

we get, $A + A' = B + B'$

and $(\alpha = ik)$, $\alpha A - \alpha A' = \kappa B - \kappa B'$

or, $2\alpha A = (\alpha + \kappa) B + (\alpha - \kappa) B'$

also, $\psi_C(L) = \psi_B(L)$; $\psi'_C(L) = \psi'_B(L)$

$$\implies Be^{\kappa L} + B'e^{-\kappa L} = Ce^{\alpha L}$$

$$\psi_A(x) = Ae^{ikx} + A'e^{-ikx}$$

$$\psi_B(x) = Be^{\kappa x} + B'e^{-\kappa x}$$

$$\psi_C(x) = Ce^{ikx}$$

using B.C. : $\psi_A(0) = \psi_B(0)$; $\psi'_A(0) = \psi'_B(0)$

we get, $A + A' = B + B'$

and $(\alpha = ik)$, $\alpha A - \alpha A' = \kappa B - \kappa B'$

or, $2\alpha A = (\alpha + \kappa) B + (\alpha - \kappa) B'$

also, $\psi_C(L) = \psi_B(L)$; $\psi'_C(L) = \psi'_B(L)$

$$\implies Be^{\kappa L} + B'e^{-\kappa L} = Ce^{\alpha L}$$

$$\text{and } \kappa Be^{\kappa L} - \kappa B'e^{-\kappa L} = \alpha Ce^{\alpha L}$$

$$\psi_A(x) = Ae^{ikx} + A'e^{-ikx}$$

$$\psi_B(x) = Be^{\kappa x} + B'e^{-\kappa x}$$

$$\psi_C(x) = Ce^{ikx}$$

using B.C. : $\psi_A(0) = \psi_B(0)$; $\psi'_A(0) = \psi'_B(0)$

we get, $A + A' = B + B'$

and $(\alpha = ik)$, $\alpha A - \alpha A' = \kappa B - \kappa B'$

or, $2\alpha A = (\alpha + \kappa) B + (\alpha - \kappa) B'$

also, $\psi_C(L) = \psi_B(L)$; $\psi'_C(L) = \psi'_B(L)$

$$\implies Be^{\kappa L} + B'e^{-\kappa L} = Ce^{\alpha L}$$

and $\kappa Be^{\kappa L} - \kappa B'e^{-\kappa L} = \alpha Ce^{\alpha L}$

write: $\eta = e^{\kappa L}$; $\beta = e^{\alpha L}$

$$\psi_A(x) = Ae^{ikx} + A'e^{-ikx}$$

$$\psi_B(x) = Be^{\kappa x} + B'e^{-\kappa x}$$

$$\psi_C(x) = Ce^{ikx}$$

using B.C. : $\psi_A(0) = \psi_B(0)$; $\psi'_A(0) = \psi'_B(0)$

we get, $A + A' = B + B'$

and $(\alpha = ik)$, $\alpha A - \alpha A' = \kappa B - \kappa B'$

or, $2\alpha A = (\alpha + \kappa) B + (\alpha - \kappa) B'$

also, $\psi_C(L) = \psi_B(L)$; $\psi'_C(L) = \psi'_B(L)$

$$\implies Be^{\kappa L} + B'e^{-\kappa L} = Ce^{\alpha L}$$

$$\text{and } \kappa Be^{\kappa L} - \kappa B'e^{-\kappa L} = \alpha Ce^{\alpha L}$$

write: $\eta = e^{\kappa L}$; $\beta = e^{\alpha L}$

$$\text{then: } \eta B + \eta^{-1} B' = \beta C$$

$$\psi_A(x) = Ae^{ikx} + A'e^{-ikx}$$

$$\psi_B(x) = Be^{\kappa x} + B'e^{-\kappa x}$$

$$\psi_C(x) = Ce^{ikx}$$

using B.C. : $\psi_A(0) = \psi_B(0)$; $\psi'_A(0) = \psi'_B(0)$

we get, $A + A' = B + B'$

and $(\alpha = ik)$, $\alpha A - \alpha A' = \kappa B - \kappa B'$

or, $2\alpha A = (\alpha + \kappa) B + (\alpha - \kappa) B'$

also, $\psi_C(L) = \psi_B(L)$; $\psi'_C(L) = \psi'_B(L)$

$\implies Be^{\kappa L} + B'e^{-\kappa L} = Ce^{\alpha L}$

and $\kappa Be^{\kappa L} - \kappa B'e^{-\kappa L} = \alpha Ce^{\alpha L}$

write: $\eta = e^{\kappa L}$; $\beta = e^{\alpha L}$

then: $\eta B + \eta^{-1} B' = \beta C$

and $\kappa \eta B - \kappa \eta^{-1} B' = \alpha \beta C$

$$\psi_A(x) = Ae^{ikx} + A'e^{-ikx}$$

$$\psi_B(x) = Be^{\kappa x} + B'e^{-\kappa x}$$

$$\psi_C(x) = Ce^{ikx}$$

using B.C. : $\psi_A(0) = \psi_B(0)$; $\psi'_A(0) = \psi'_B(0)$

we get, $A + A' = B + B'$

and $(\alpha = ik)$, $\alpha A - \alpha A' = \kappa B - \kappa B'$

or, $2\alpha A = (\alpha + \kappa) B + (\alpha - \kappa) B'$

also, $\psi_C(L) = \psi_B(L)$; $\psi'_C(L) = \psi'_B(L)$

$$\implies Be^{\kappa L} + B'e^{-\kappa L} = Ce^{\alpha L}$$

$$\text{and } \kappa Be^{\kappa L} - \kappa B'e^{-\kappa L} = \alpha Ce^{\alpha L}$$

$$\text{write: } \eta = e^{\kappa L}; \beta = e^{\alpha L}$$

$$\text{then: } \eta B + \eta^{-1} B = \beta C$$

$$\text{and } \kappa \eta B - \kappa \eta^{-1} B = \alpha \beta C$$

$$\text{get: } 2\kappa \eta B = \beta C(\kappa + \alpha)$$

$$\psi_A(x) = Ae^{ikx} + A'e^{-ikx}$$

$$\psi_B(x) = Be^{\kappa x} + B'e^{-\kappa x}$$

$$\psi_C(x) = Ce^{ikx}$$

using B.C. : $\psi_A(0) = \psi_B(0)$; $\psi'_A(0) = \psi'_B(0)$

we get, $A + A' = B + B'$

and $(\alpha = ik)$, $\alpha A - \alpha A' = \kappa B - \kappa B'$

or, $2\alpha A = (\alpha + \kappa) B + (\alpha - \kappa) B'$

also, $\psi_C(L) = \psi_B(L)$; $\psi'_C(L) = \psi'_B(L)$

$$\implies Be^{\kappa L} + B'e^{-\kappa L} = Ce^{\alpha L}$$

$$\text{and } \kappa Be^{\kappa L} - \kappa B'e^{-\kappa L} = \alpha Ce^{\alpha L}$$

$$\text{write: } \eta = e^{\kappa L}; \beta = e^{\alpha L}$$

$$\text{then: } \eta B + \eta^{-1} B' = \beta C$$

$$\text{and } \kappa \eta B - \kappa \eta^{-1} B' = \alpha \beta C$$

$$\text{get: } 2\kappa \eta B = \beta C(\kappa + \alpha)$$

$$\text{and } 2\kappa \eta^{-1} B' = \beta C(\kappa - \alpha)$$

$$\psi_A(x) = Ae^{ikx} + A'e^{-ikx}$$

$$\psi_B(x) = Be^{\kappa x} + B'e^{-\kappa x}$$

$$\psi_C(x) = Ce^{ikx}$$

using B.C. : $\psi_A(0) = \psi_B(0)$; $\psi'_A(0) = \psi'_B(0)$

we get, $A + A' = B + B'$

and $(\alpha = ik)$, $\alpha A - \alpha A' = \kappa B - \kappa B'$

or, $2\alpha A = (\alpha + \kappa) B + (\alpha - \kappa) B'$

also, $\psi_C(L) = \psi_B(L)$; $\psi'_C(L) = \psi'_B(L)$

$$\implies Be^{\kappa L} + B'e^{-\kappa L} = Ce^{\alpha L}$$

$$\text{and } \kappa Be^{\kappa L} - \kappa B'e^{-\kappa L} = \alpha Ce^{\alpha L}$$

$$\text{write: } \eta = e^{\kappa L}; \beta = e^{\alpha L}$$

$$\text{then: } \eta B + \eta^{-1} B = \beta C$$

$$\text{and } \kappa \eta B - \kappa \eta^{-1} B = \alpha \beta C$$

$$\text{get: } 2\kappa \eta B = \beta C(\kappa + \alpha)$$

$$\text{and } 2\kappa \eta^{-1} B' = \beta C(\kappa - \alpha)$$

Task: find tunneling probability

$$\psi_A(x) = Ae^{ikx} + A'e^{-ikx}$$

$$\psi_B(x) = Be^{\kappa x} + B'e^{-\kappa x}$$

$$\psi_C(x) = Ce^{ikx}$$

using B.C. : $\psi_A(0) = \psi_B(0)$; $\psi'_A(0) = \psi'_B(0)$

we get, $A + A' = B + B'$

and $(\alpha = ik)$, $\alpha A - \alpha A' = \kappa B - \kappa B'$

or, $2\alpha A = (\alpha + \kappa) B + (\alpha - \kappa) B'$

also, $\psi_C(L) = \psi_B(L)$; $\psi'_C(L) = \psi'_B(L)$

$$\implies Be^{\kappa L} + B'e^{-\kappa L} = Ce^{\alpha L}$$

$$\text{and } \kappa Be^{\kappa L} - \kappa B'e^{-\kappa L} = \alpha Ce^{\alpha L}$$

$$\text{write: } \eta = e^{\kappa L}; \beta = e^{\alpha L}$$

$$\text{then: } \eta B + \eta^{-1} B' = \beta C$$

$$\text{and } \kappa \eta B - \kappa \eta^{-1} B' = \alpha \beta C$$

$$\text{get: } 2\kappa \eta B = \beta C(\kappa + \alpha)$$

$$\text{and } 2\kappa \eta^{-1} B' = \beta C(\kappa - \alpha)$$

Task: find tunneling probability

$$= \frac{\text{prob. of particle moving rightward in Zone C}}{\text{prob. of particle moving rightward in Zone A}}$$

$$\psi_A(x) = Ae^{ikx} + A'e^{-ikx}$$

$$\psi_B(x) = Be^{\kappa x} + B'e^{-\kappa x}$$

$$\psi_C(x) = Ce^{ikx}$$

using B.C. : $\psi_A(0) = \psi_B(0)$; $\psi'_A(0) = \psi'_B(0)$

we get, $A + A' = B + B'$

and $(\alpha = ik)$, $\alpha A - \alpha A' = \kappa B - \kappa B'$

or, $2\alpha A = (\alpha + \kappa) B + (\alpha - \kappa) B'$

also, $\psi_C(L) = \psi_B(L)$; $\psi'_C(L) = \psi'_B(L)$

$$\implies Be^{\kappa L} + B'e^{-\kappa L} = Ce^{\alpha L}$$

$$\text{and } \kappa Be^{\kappa L} - \kappa B'e^{-\kappa L} = \alpha Ce^{\alpha L}$$

$$\text{write: } \eta = e^{\kappa L}; \beta = e^{\alpha L}$$

$$\text{then: } \eta B + \eta^{-1} B' = \beta C$$

$$\text{and } \kappa \eta B - \kappa \eta^{-1} B' = \alpha \beta C$$

$$\text{get: } 2\kappa \eta B = \beta C(\kappa + \alpha)$$

$$\text{and } 2\kappa \eta^{-1} B' = \beta C(\kappa - \alpha)$$

Task: find tunneling probability

$$= \frac{\text{prob. of particle moving rightward in Zone C}}{\text{prob. of particle moving rightward in Zone A}} = \frac{|C|^2}{|A|^2}$$

$$\psi_A(x) = Ae^{ikx} + A'e^{-ikx}$$

$$\psi_B(x) = Be^{\kappa x} + B'e^{-\kappa x}$$

$$\psi_C(x) = Ce^{ikx}$$

using B.C. : $\psi_A(0) = \psi_B(0)$; $\psi'_A(0) = \psi'_B(0)$

we get, $A + A' = B + B'$

and $(\alpha = ik)$, $\alpha A - \alpha A' = \kappa B - \kappa B'$

or, $2\alpha A = (\alpha + \kappa)B + (\alpha - \kappa)B'$

also, $\psi_C(L) = \psi_B(L)$; $\psi'_C(L) = \psi'_B(L)$

$$\implies Be^{\kappa L} + B'e^{-\kappa L} = Ce^{\alpha L}$$

$$\text{and } \kappa Be^{\kappa L} - \kappa B'e^{-\kappa L} = \alpha Ce^{\alpha L}$$

$$\text{write: } \eta = e^{\kappa L}; \beta = e^{\alpha L}$$

$$\text{then: } \eta B + \eta^{-1}B' = \beta C$$

$$\text{and } \kappa\eta B - \kappa\eta^{-1}B' = \alpha\beta C$$

$$\text{get: } 2\kappa\eta B = \beta C(\kappa + \alpha)$$

$$\text{and } 2\kappa\eta^{-1}B' = \beta C(\kappa - \alpha)$$

Task: find tunneling probability

$$= \frac{\text{prob. of particle moving rightward in Zone C}}{\text{prob. of particle moving rightward in Zone A}} = \frac{|C|^2}{|A|^2}$$

i.e., we need C in terms of A

$$2\alpha A = (\alpha + \kappa) B + (\alpha - \kappa) B'$$

$$2\kappa\eta B = \beta C(\kappa + \alpha)$$

$$2\kappa\eta^{-1}B' = \beta C(\kappa - \alpha)$$

$$\begin{array}{l|l}
 2\alpha A = (\alpha + \kappa) B + (\alpha - \kappa) B' & 2\alpha A = (\alpha + \kappa) \frac{\beta C(\kappa + \alpha)}{2\kappa\eta} + (\alpha - \kappa) \frac{\beta C(\kappa - \alpha)}{2\kappa\eta^{-1}} \\
 2\kappa\eta B = \beta C(\kappa + \alpha) & \\
 2\kappa\eta^{-1} B' = \beta C(\kappa - \alpha) &
 \end{array}$$

$$\begin{array}{l|l}
2\alpha A = (\alpha + \kappa) B + (\alpha - \kappa) B' & 2\alpha A = (\alpha + \kappa) \frac{\beta C(\kappa + \alpha)}{2\kappa\eta} + (\alpha - \kappa) \frac{\beta C(\kappa - \alpha)}{2\kappa\eta^{-1}} \\
2\kappa\eta B = \beta C(\kappa + \alpha) & = \frac{\beta C}{2\kappa} \left[\frac{(\alpha + \kappa)^2}{\eta} - \frac{(\alpha - \kappa)^2}{\eta^{-1}} \right] \\
2\kappa\eta^{-1} B' = \beta C(\kappa - \alpha) &
\end{array}$$

$$\begin{array}{l|l}
2\alpha A = (\alpha + \kappa) B + (\alpha - \kappa) B' & 2\alpha A = (\alpha + \kappa) \frac{\beta C(\kappa + \alpha)}{2\kappa\eta} + (\alpha - \kappa) \frac{\beta C(\kappa - \alpha)}{2\kappa\eta^{-1}} \\
2\kappa\eta B = \beta C(\kappa + \alpha) & = \frac{\beta C}{2\kappa} \left[\frac{(\alpha + \kappa)^2}{\eta} - \frac{(\alpha - \kappa)^2}{\eta^{-1}} \right] \\
2\kappa\eta^{-1} B' = \beta C(\kappa - \alpha) & \alpha = ik; \quad \eta = e^{\kappa L}; \quad \beta = e^{\alpha L}
\end{array}$$

$$\begin{array}{l|l}
2\alpha A = (\alpha + \kappa) B + (\alpha - \kappa) B' & 2\alpha A = (\alpha + \kappa) \frac{\beta C(\kappa + \alpha)}{2\kappa\eta} + (\alpha - \kappa) \frac{\beta C(\kappa - \alpha)}{2\kappa\eta^{-1}} \\
2\kappa\eta B = \beta C(\kappa + \alpha) & = \frac{\beta C}{2\kappa} \left[\frac{(\alpha + \kappa)^2}{\eta} - \frac{(\alpha - \kappa)^2}{\eta^{-1}} \right] \\
2\kappa\eta^{-1} B' = \beta C(\kappa - \alpha) & \alpha = ik; \eta = e^{\kappa L}; \beta = e^{\alpha L}
\end{array}$$

$$\therefore \frac{C}{A} = \frac{4\alpha\kappa}{\beta \left[\frac{(\alpha + \kappa)^2}{\eta} - \frac{(\alpha - \kappa)^2}{\eta^{-1}} \right]}$$

$$\begin{array}{l|l}
2\alpha A = (\alpha + \kappa) B + (\alpha - \kappa) B' & 2\alpha A = (\alpha + \kappa) \frac{\beta C(\kappa + \alpha)}{2\kappa\eta} + (\alpha - \kappa) \frac{\beta C(\kappa - \alpha)}{2\kappa\eta^{-1}} \\
2\kappa\eta B = \beta C(\kappa + \alpha) & = \frac{\beta C}{2\kappa} \left[\frac{(\alpha + \kappa)^2}{\eta} - \frac{(\alpha - \kappa)^2}{\eta^{-1}} \right] \\
2\kappa\eta^{-1} B' = \beta C(\kappa - \alpha) & \alpha = ik; \eta = e^{\kappa L}; \beta = e^{\alpha L}
\end{array}$$

$$\therefore \frac{C}{A} = \frac{4\alpha\kappa}{\beta \left[\frac{(\alpha + \kappa)^2}{\eta} - \frac{(\alpha - \kappa)^2}{\eta^{-1}} \right]} \stackrel{\alpha = ik}{=} -\frac{1}{\beta} \frac{4i\kappa k}{(\eta - \eta^{-1})(\kappa^2 - k^2) + 2i\kappa k(\eta + \eta^{-1})}$$

$$\begin{array}{l|l}
2\alpha A = (\alpha + \kappa) B + (\alpha - \kappa) B' & 2\alpha A = (\alpha + \kappa) \frac{\beta C(\kappa + \alpha)}{2\kappa\eta} + (\alpha - \kappa) \frac{\beta C(\kappa - \alpha)}{2\kappa\eta^{-1}} \\
2\kappa\eta B = \beta C(\kappa + \alpha) & = \frac{\beta C}{2\kappa} \left[\frac{(\alpha + \kappa)^2}{\eta} - \frac{(\alpha - \kappa)^2}{\eta^{-1}} \right] \\
2\kappa\eta^{-1} B' = \beta C(\kappa - \alpha) & \alpha = ik; \eta = e^{\kappa L}; \beta = e^{\alpha L}
\end{array}$$

$$\therefore \frac{C}{A} = \frac{4\alpha\kappa}{\beta \left[\frac{(\alpha + \kappa)^2}{\eta} - \frac{(\alpha - \kappa)^2}{\eta^{-1}} \right]} \stackrel{\alpha = ik}{=} -\frac{1}{\beta} \frac{4i\kappa k}{(\eta - \eta^{-1})(\kappa^2 - k^2) + 2i\kappa k(\eta + \eta^{-1})}$$

$$\therefore \frac{|C|^2}{|A|^2} \stackrel{|\beta|^2=1}{=} \frac{16\kappa^2 k^2}{(\eta - \eta^{-1})^2 (\kappa^2 - k^2)^2 + 4\kappa^2 k^2 (\eta + \eta^{-1})^2}$$

$$\begin{array}{l|l}
2\alpha A = (\alpha + \kappa) B + (\alpha - \kappa) B' & 2\alpha A = (\alpha + \kappa) \frac{\beta C(\kappa + \alpha)}{2\kappa\eta} + (\alpha - \kappa) \frac{\beta C(\kappa - \alpha)}{2\kappa\eta^{-1}} \\
2\kappa\eta B = \beta C(\kappa + \alpha) & = \frac{\beta C}{2\kappa} \left[\frac{(\alpha + \kappa)^2}{\eta} - \frac{(\alpha - \kappa)^2}{\eta^{-1}} \right] \\
2\kappa\eta^{-1} B' = \beta C(\kappa - \alpha) & \alpha = ik; \eta = e^{\kappa L}; \beta = e^{\alpha L}
\end{array}$$

$$\therefore \frac{C}{A} = \frac{4\alpha\kappa}{\beta \left[\frac{(\alpha + \kappa)^2}{\eta} - \frac{(\alpha - \kappa)^2}{\eta^{-1}} \right]} \stackrel{\alpha = ik}{=} -\frac{1}{\beta} \frac{4i\kappa k}{(\eta - \eta^{-1})(\kappa^2 - k^2) + 2i\kappa k(\eta + \eta^{-1})}$$

$$\therefore \frac{|C|^2}{|A|^2} \stackrel{|\beta|^2=1}{=} \frac{16\kappa^2 k^2}{(\eta - \eta^{-1})^2 (\kappa^2 - k^2)^2 + 4\kappa^2 k^2 (\eta + \eta^{-1})^2} = \frac{16\kappa^2 k^2}{(\eta - \eta^{-1})^2 (\kappa^2 - k^2)^2 + 4\kappa^2 k^2 [(\eta - \eta^{-1})^2 + 4]}$$

$$\begin{array}{l|l}
2\alpha A = (\alpha + \kappa) B + (\alpha - \kappa) B' & 2\alpha A = (\alpha + \kappa) \frac{\beta C(\kappa + \alpha)}{2\kappa\eta} + (\alpha - \kappa) \frac{\beta C(\kappa - \alpha)}{2\kappa\eta^{-1}} \\
2\kappa\eta B = \beta C(\kappa + \alpha) & = \frac{\beta C}{2\kappa} \left[\frac{(\alpha + \kappa)^2}{\eta} - \frac{(\alpha - \kappa)^2}{\eta^{-1}} \right] \\
2\kappa\eta^{-1} B' = \beta C(\kappa - \alpha) & \alpha = ik; \eta = e^{\kappa L}; \beta = e^{\alpha L}
\end{array}$$

$$\therefore \frac{C}{A} = \frac{4\alpha\kappa}{\beta \left[\frac{(\alpha + \kappa)^2}{\eta} - \frac{(\alpha - \kappa)^2}{\eta^{-1}} \right]} \stackrel{\alpha = ik}{=} -\frac{1}{\beta} \frac{4i\kappa k}{(\eta - \eta^{-1})(\kappa^2 - k^2) + 2i\kappa k(\eta + \eta^{-1})}$$

$$\begin{aligned}
\therefore \frac{|C|^2}{|A|^2} & \stackrel{|\beta|^2=1}{=} \frac{16\kappa^2 k^2}{(\eta - \eta^{-1})^2 (\kappa^2 - k^2)^2 + 4\kappa^2 k^2 (\eta + \eta^{-1})^2} = \frac{16\kappa^2 k^2}{(\eta - \eta^{-1})^2 (\kappa^2 - k^2)^2 + 4\kappa^2 k^2 [(\eta - \eta^{-1})^2 + 4]} \\
& = \frac{16\kappa^2 k^2}{(\eta - \eta^{-1})^2 [(\kappa^2 - k^2)^2 + 4\kappa^2 k^2] + 16\kappa^2 k^2}
\end{aligned}$$

$$\begin{array}{l|l}
2\alpha A = (\alpha + \kappa) B + (\alpha - \kappa) B' & 2\alpha A = (\alpha + \kappa) \frac{\beta C(\kappa + \alpha)}{2\kappa\eta} + (\alpha - \kappa) \frac{\beta C(\kappa - \alpha)}{2\kappa\eta^{-1}} \\
2\kappa\eta B = \beta C(\kappa + \alpha) & = \frac{\beta C}{2\kappa} \left[\frac{(\alpha + \kappa)^2}{\eta} - \frac{(\alpha - \kappa)^2}{\eta^{-1}} \right] \\
2\kappa\eta^{-1} B' = \beta C(\kappa - \alpha) & \alpha = ik; \eta = e^{\kappa L}; \beta = e^{\alpha L}
\end{array}$$

$$\therefore \frac{C}{A} = \frac{4\alpha\kappa}{\beta \left[\frac{(\alpha + \kappa)^2}{\eta} - \frac{(\alpha - \kappa)^2}{\eta^{-1}} \right]} \stackrel{\alpha = ik}{=} -\frac{1}{\beta} \frac{4i\kappa k}{(\eta - \eta^{-1})(\kappa^2 - k^2) + 2i\kappa k(\eta + \eta^{-1})}$$

$$\begin{aligned}
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$$\begin{array}{l|l}
2\alpha A = (\alpha + \kappa) B + (\alpha - \kappa) B' & 2\alpha A = (\alpha + \kappa) \frac{\beta C(\kappa + \alpha)}{2\kappa\eta} + (\alpha - \kappa) \frac{\beta C(\kappa - \alpha)}{2\kappa\eta^{-1}} \\
2\kappa\eta B = \beta C(\kappa + \alpha) & = \frac{\beta C}{2\kappa} \left[\frac{(\alpha + \kappa)^2}{\eta} - \frac{(\alpha - \kappa)^2}{\eta^{-1}} \right] \\
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\end{aligned}$$

or, $\frac{|C|^2}{|A|^2} = \frac{1}{1+G}$, where $G = \frac{(\eta - \eta^{-1})^2}{\frac{16\kappa^2 k^2}{(\kappa^2 + k^2)^2}}$

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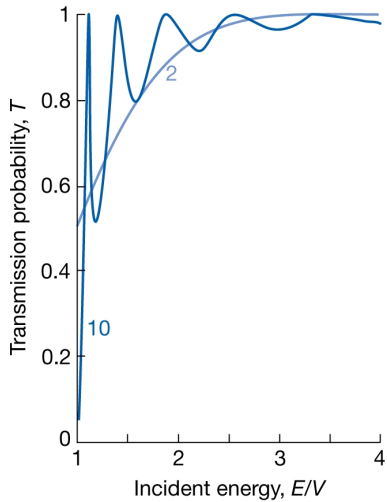
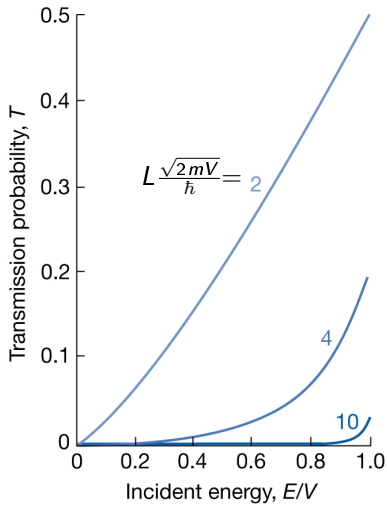
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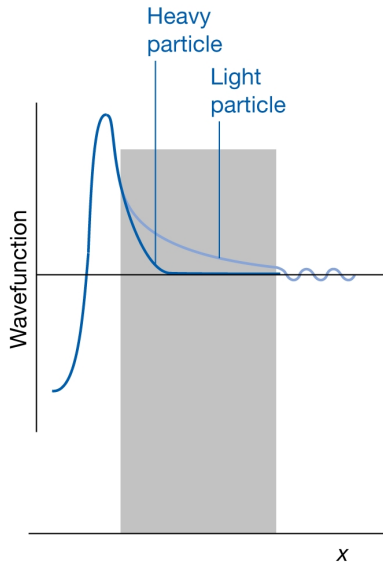
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$$\text{Transmission prob.} = \frac{1}{1+G}$$





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$\psi = \phi_1 \phi_2$ is a solution of the SE with eigenvalue, $E = \epsilon_1 + \epsilon_2$

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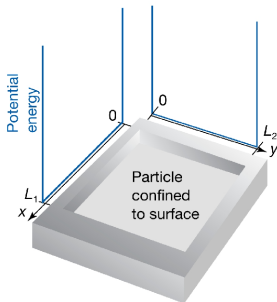
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$$V(x, y) = V_x(x) + V_y(y); \text{ where } V_x(x) = 0 \quad \forall 0 < x < L_x \text{ and } \infty \text{ elsewhere; } V_y(y)$$

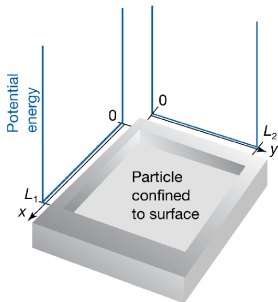


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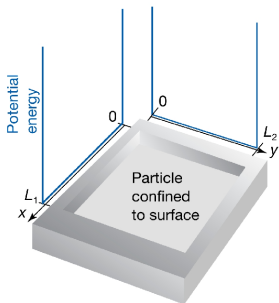
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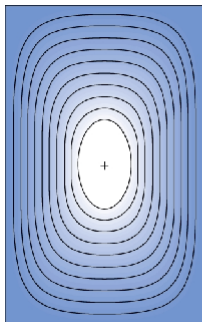
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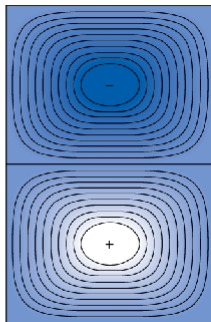
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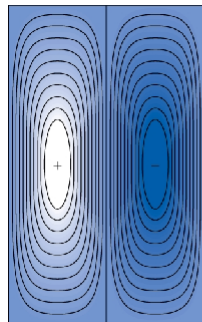
eigenstates depicted as contour diagrams:



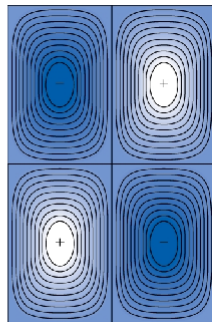
(a)



(b)

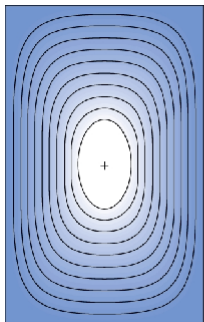


(c)

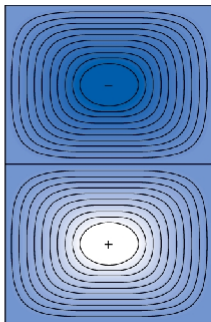


(d)

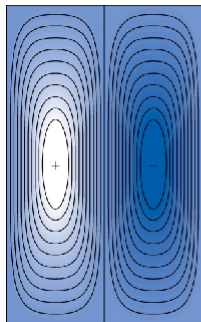
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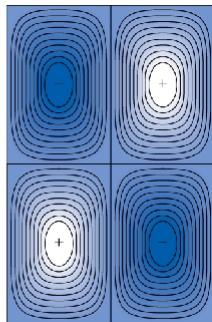
(a)



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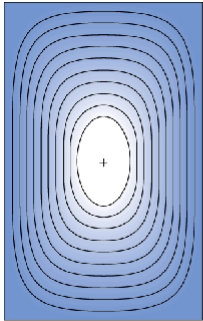
(c)



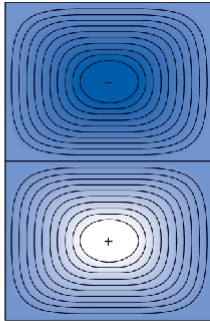
(d)

Identify quantum numbers corresponding to the four diagrams shown.

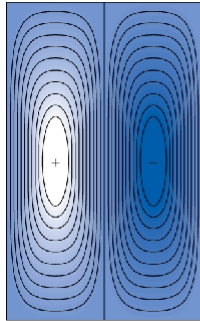
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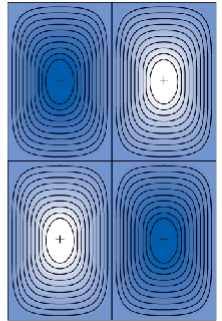
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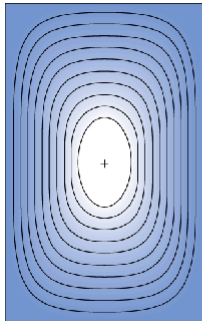
(c)



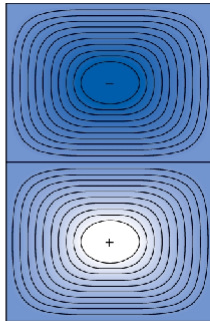
(d)

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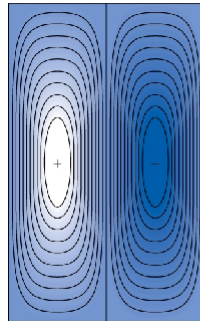
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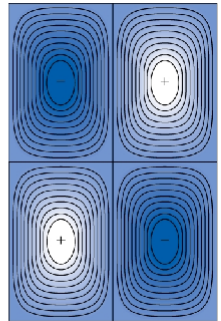
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Will these be there if we plot the densities instead of the wavefunctions?

generalise to higher dimensions :

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where H_{v_i} are the Hermite polynomials, $H_0(x) = 1$; $H_1(x) = 2x$ and so on;

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Extend to spherical oscillator in 3-D

